Camila Eloise Lightfoot

U.S. Citizen

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Education

B.S. in Computer Science — Expected Graduation Winter 2025
George Mason University, Fairfax, VA
Overall GPA: **3.52/4.00**

- Relevant Courses: Database Concepts, Mobile Applications Development, Computer Systems Programming, Software Engineering, Introduction to Low-Level Programming, Secure programming and Systems.

Certification in Machine Learning Foundations — May 2025
eCornell University program
Overall GPA: **4.00/4.00**

A.S. in Computer Science — Aug 2021 - May 2023
Northern Virginia Community College, loudoun, VA
Overall GPA: **4.00/4.00**

- Relevant courses: Object-Oriented Programming, Data Structures & Analysis of Algorithms

Experience

Instructor, CodeAdvantage, Vienna, VA — February 2024 - December 2024

- Teaching young students foundational computer science concepts and programming using Scratch and ScratchJr.

Fellow, Break Through Tech AI Studio, VA *GitHub: My eCornell Portfolio Repo + README* — May 2025 - Present

- Completed Break Through Tech AI's Foundations of Machine Learning program, building hands-on projects with Python and scikit-learn, applying key ML techniques (classification, clustering, neural networks), and exploring responsible AI practices.

Projects

Planned Parenthood AI Studio Project *GitHub: PPFA AI Studio 2025 Repo + README* — May 2025

- Collaborating with a team through Break Through Tech AI Studio to enhance Planned Parenthood's SRH chatbot.
- Designing and training machine learning models for input/output classification to improve escalation to human educators.
- Conducting data cleaning, augmentation, and evaluation to address false negatives and true negatives.

Process Management and System Calls — Spring 2025

- Extended kernel to support fork(), waitpid(), and process lifecycle management.
- Implemented trapframe copying and address space duplication for new processes.
- Fixed PID reuse issues, synchronized parent-child execution, and passed at-fork-limit and waitpid tests.

Customer and Order Management System — November 2024

- Designed and migrated a relational database from Oracle SQL to MongoDB using an embedded document model. Developed SQL scripts for data conversion, and built a Java-based CLI with JDBC for database management.

Otur CPU Scheduler Implementation — February 2024

- Implemented core functions for the Otur CPU Scheduler in the StrawHat Virtual Machine using C, simulating CPU scheduling with multi-level feedback queue algorithms.
- Managed process states using bitwise operations, handling process queues.

Technical Skills

Programming Skills	Java, C, Python, JavaScript, NoSQL (e.g., MongoDB), SQL, Git
Operating Systems	Windows OS, Android OS, Linux/Unix
Libraries/Frameworks/Tools	Pandas, NumPy, scikit-learn, Flask, GitHub, Jupyter, VS Code, Word, Excel, PowerPoint
Languages	English, Spanish

Honors & Interests

Honors	Grace Hopper Conference 2023 - Break Through Tech Mason Scholar Awarded by AnitaB.org
Interests	Database Design & Management, Web Development, Data Science, Machine Learning